Cross-Dataset Pruning Study: ResNet18 and VGG-16 on Optical (FloodNet) vs. SAR (Sen1Floods11) Flood Classification

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Setup

Both architectures were trained with an identical supervised pipeline on both datasets: image-level binary flood classification, 50/50 stratified train/validation split (seed 42), 400×300 inputs, horizontal/vertical flip augmentation, class-balanced sampling, SGD (lr $= 0.01$), 50 dense epochs. Sen1Floods11 is per-pixel labeled, so chips were converted to image-level labels (flooded if $\geq 1\%$ of valid pixels are water; 441 chips, 293 flooded). FloodNet uses its 398 labeled RGB images (51 flooded). Each model was pruned with global L1 *unstructured* and L1 *structured* (filter/neuron) magnitude pruning at $\{10, 30, 50, 70, 90\}\%$, then fine-tuned for 50 epochs. Reported numbers are best-validation-F1 checkpoints.

Table 1: Full results. Params = nonzero parameters after pruning. Red rows are degenerate collapses: the network predicts "flooded" for every image (recall = 1, AUC ≈ 0.5).

				FloodNet (optical)		**Sen1Floods11 (SAR)**	
Arch.	**Method**	**Ratio**	**Params (M)**	**Acc.**	**F1**	**Acc.**	**F1**
ResNet18	dense	0%	11.18	0.975	0.898	0.878	0.907
	unstructured	10%	10.06	0.975	0.898	0.873	0.902
	unstructured	30%	7.83	0.975	0.898	0.891	0.918
	unstructured	50%	5.59	0.970	0.875	0.873	0.902
	unstructured	70%	3.36	0.975	0.902	0.873	0.904
	unstructured	90%	1.13	0.965	0.857	0.833	0.874
	structured	10%	10.06	0.970	0.880	0.842	0.881
	structured	30%	7.82	0.985	0.939	0.842	0.879
	structured	50%	5.59	0.975	0.902	0.828	0.873
	structured	70%	3.37	0.970	0.875	0.801	0.853
	structured	90%	1.13	0.930	0.750	0.697	0.806
VGG-16	dense	0%	134.27	0.970	0.875	0.824	0.864
	unstructured	10%	120.84	0.975	0.902	0.824	0.865
	unstructured	30%	93.99	0.975	0.898	0.819	0.865
	unstructured	50%	67.14	0.980	0.917	0.819	0.865
	unstructured	70%	40.29	0.980	0.917	0.824	0.867
	unstructured	90%	13.44	0.980	0.923	0.824	0.864
	structured	10%	120.83	0.975	0.898	0.828	0.868
	structured	30%	93.98	0.980	0.926	0.824	0.869
	structured	50%	67.14	0.975	0.894	0.747	0.833
	structured	70%	40.31	0.960	0.833	0.665	0.799
	structured	90%	13.45	0.126	0.223	0.665	0.799

Key findings

1. **Unstructured pruning holds on BOTH modalities.** Up to 90% sparsity, F1 stays within ± 0.05 of dense for both architectures on both datasets. Notably, VGG-16 at 90% unstructured (13.4M nonzero params, ResNet18 scale) *exceeds* its dense baseline on FloodNet (0.923 vs. 0.875) and matches it on SAR (0.864).

2. **Structured pruning is where behaviour diverges — and SAR breaks first.** VGG-16 collapses at 70% structured on SAR but only at 90% on optical. ResNet18 never fully collapses, but on SAR it degrades monotonically from the very first structured level (-0.026 F1 at 10%, -0.101 at 90%, accuracy $0.878 \rightarrow 0.697$), whereas on optical it is stable until 70% and only drops at 90%.

3. **The collapse mode is a majority-class predictor, and class balance can hide it.** Collapsed runs predict "flooded" everywhere (recall = 1.0, AUC ≈ 0.5). Since Sen1Floods11 is 66% flooded, a collapsed model still shows F1 ≈ 0.80 there — the F1 alone understates the SAR failure; on FloodNet (13% flooded) the same failure is obvious (F1 0.22). Accuracy and AUC should accompany F1 in all future comparisons.

Interpretation and proposed next steps

The answer to our question is the more interesting case: a heavily pruned classic CNN keeps its accuracy on both optical and SAR imagery *only* under unstructured sparsity; under structured (filter-level) pruning the SAR task degrades earlier and more steeply than the optical one, on both architectures. A plausible explanation is that structured pruning removes entire ImageNet-pretrained filters, and the SAR task — already out-of-distribution for these filters — depends on a small subset of low-level texture channels that magnitude ranking does not protect; unstructured pruning preserves fragments of every filter. Proposed follow-ups: (i) repeat the structured runs over 2–3 seeds to confirm the monotonic SAR degradation is not noise; (ii) channel-importance analysis of which filters are removed on SAR vs. optical; (iii) test gentler fine-tuning (lower LR, longer schedule) and iterative/gradual structured pruning to see whether the SAR collapse is recoverable.